RCC BEAM DESIGNER

IS 456:2000 — LIMIT STATE DESIGN

Project Name:	RCC Beam Design Calculation	Date:	Auto-generated
Design Code:	IS 456:2000	Units:	mm, kN, N/mm²
Report Type:	Complete Report	Revision:	0

1. DESIGN INPUT SUMMARY

GEOMETRY		MATERIAL	
Beam Width (b)	300 mm	**Concrete Grade**	M25
Overall Depth (D)	500 mm	**fck**	25 N/mm²
Effective Cover	40 mm	**Steel Grade**	Fe500
Span (L)	5.00 m	**fy**	500 N/mm²
Support Condition	Simply Supported		
DESIGN LOADS			
Factored Moment (Mu)	180.00 kN·m	Factored Shear (Vu)	120.00 kN

2. FLEXURAL DESIGN

2.1 Effective Depth

d = D - effective cover = 500.0 - 40.0 = **460.0 mm**

2.2 Limiting Neutral Axis & Moment

$x_{u,max}$ = **211.60 mm**
M_{lim} = **212.03 kN·m**

2.3 Section Classification

M_u (180.00) ≤ M_{lim} (212.03) → **SINGLY REINFORCED**

3. SHEAR DESIGN

3.1 Shear Stress Checks

τ_v = **0.870 N/mm²**
τ_c = **0.615 N/mm²** (Table 19)
$\tau_{c,max}$ = **3.10 N/mm²** (Table 20)

3.2 Shear Reinforcement

$\tau_v > \tau_c$, required. Provided: **8 mm Ø, 2-legged @ 300 mm c/c**

4. SERVICEABILITY CHECK

Basic Ratio = 20, Mod Factor = 0.960

Allowable Ratio = **19.20**, Actual L/d = **10.87**

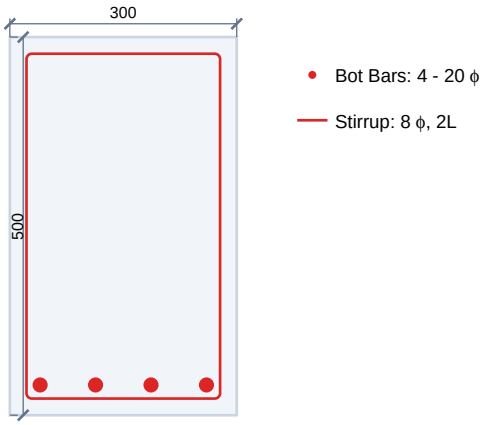
✓ **SAFE IN SERVICEABILITY**

5. DETAILING CHECK

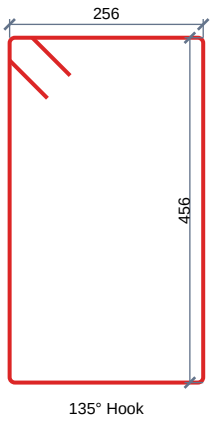
Minimum Ast Required (Cl. 26.5.1)	Provided 1256.6 ≥ 234.6 mm² — OK
Maximum Ast Allowed (4% gross)	Provided 1256.6 ≤ 6000.0 mm² — OK
Neutral Axis of Provided Steel (xu ≤ xu,max)	202.5 ≤ 211.6 mm — OK
Minimum Clear Spacing	25.0 mm
Actual Clear Spacing	53.3 mm — OK
Development Length (Ld)	971.0 mm
Clear Cover to Links (Cl. 26.4.1)	22.0 ≥ 20.0 mm — OK

6. ENGINEERING DRAWINGS

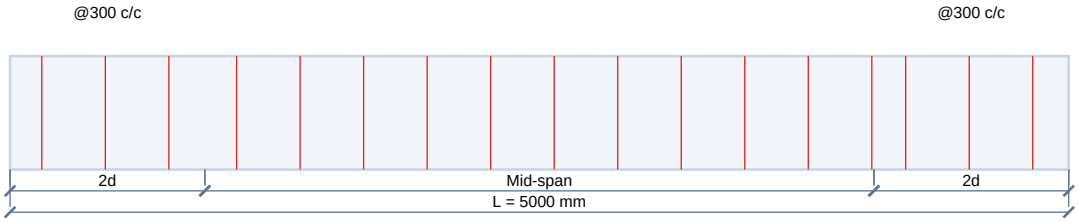
CROSS SECTION (END VIEW)



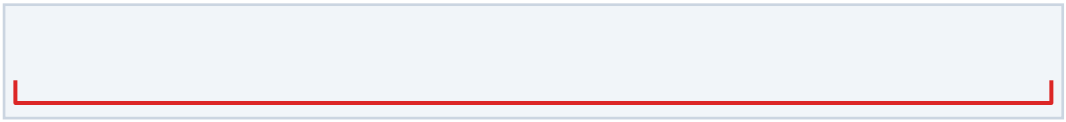
STIRRUP DETAIL



STIRRUP SPACING DETAIL (SIDE VIEW)



LONGITUDINAL SECTION



4 - 20 ϕ (Bot)

7. DETAILED CALCULATION STEPS

7.1 Flexural Design (Clause 38)

1. Effective Depth (d):

d = D - effective_cover

d = 500.0 - 40.0 = **460.00 mm**

2. Limiting Neutral Axis ($x_{u,max}$):

$x_{u,max}$ = 0.46 × d = 0.46 × 460.00 = **211.60 mm**

3. Limiting Moment Capacity (M_{lim}):

$M_{lim} = 0.36 \times f_{ck} \times b \times x_{u,max} \times (d - 0.42 \times x_{u,max}) / 10^6$

M_{lim} = 0.36 × 25 × 300.0 × 211.60 × (460.00 - 0.42 × 211.60) / 10^6 = **212.03 kN·m**

4. Required Tension Steel (A_{st}):

$A_{st} = (0.5 \times f_{ck} / f_y) \times [1 - \sqrt{(1 - 4.6 \times M_u \times 10^6 / (f_{ck} \times b \times d^2))}] \times b \times d$

A_{st} = (0.5 × 25 / 500) × [1 - √(1 - 4.6 × 180000000 / (25 × 300.0 × 460.00^2))] × 300.0 × 460.00

A_{st} = 1064.1 mm²

7.2 Shear Design (Clause 40)

1. Nominal Shear Stress (τ_v):

$\tau_v = (V_u \times 1000) / (b \times d)$

τ_v = (120.0 × 1000) / (300.0 × 460.00) = **0.870 N/mm²**

2. Percentage of Tension Steel (P_t):

$P_t = (100 \times A_{st}) / (b \times d)$

P_t = (100 × 1256.6) / (300.0 × 460.00) = **0.911 %**

3. Design Concrete Shear Strength (τ_c):

Interpolated from IS 456 Table 19 for M25 and P_t = 0.911%

τ_c = 0.615 N/mm²

4. Shear Reinforcement Spacing (s):

$V_{us} = (\tau_v - \tau_c) \times b \times d / 1000$

$s = (0.87 \times f_y \times A_{sv} \times d) / (V_{us} \times 1000)$

Provided Spacing = 300 mm c/c

7.3 Serviceability Check (Clause 23.2)

1. Actual Steel Stress (f_s):

$f_s = 0.58 \times f_y \times (A_{st,req} / A_{st,prov})$

$f_s = 0.58 \times 500 \times (1064.1 / 1256.6) =$ **245.57 N/mm²**

2. Modification Factor (K_t):

From a published curve-fit of IS 456 Fig. 4 for $f_s = 245.57$ and $P_t = 0.911\%$ (provided steel)

$K_t = 0.960$

3. Allowable Span/Depth Ratio:

Allowable = Basic Ratio × K_t = 20.00 × 0.960 = **19.20**

4. Actual Span/Depth Ratio:

Actual = (Span × 1000) / d = (5.0 × 1000) / 460.00 = **10.87**

7. SUMMARY & REPORT

Design Check	Result	Requirement	Status
Flexural Design	Mu = 180.0 kN·m	Mlim = 212.0 kN·m	SAFE
Shear Design	Vu = 120.0 kN	$\tau v \leq \tau c,max$	SAFE
Serviceability	L/d = 10.9	Allowable = 19.2	SAFE
Detailing	Per IS 456 Cl. 26	—	ADEQUATE

OVERALL DESIGN STATUS: ✓ DESIGN ACCEPTABLE

8. FINAL REINFORCEMENT SUMMARY

Beam: 300 × 500 mm

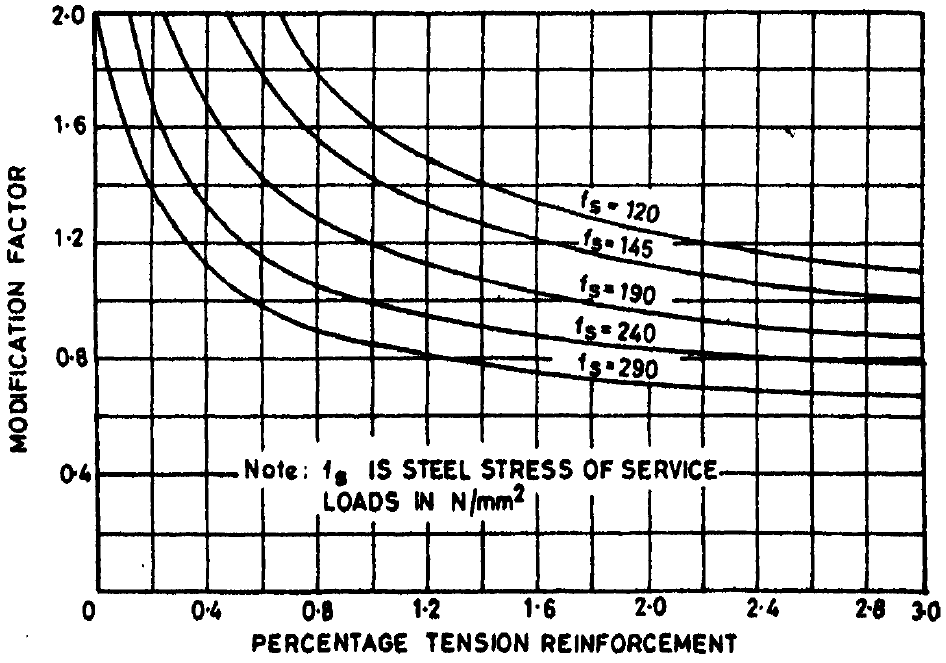
Bottom (Tension): 4 × 20 mm Ø

Stirrups:
8 mm Ø, 2-legged @ 300 mm c/c (within 920 mm of supports)
8 mm Ø, 2-legged @ 300 mm c/c (in remaining mid-span)

Clear Cover to Links: 22.0 mm

APPENDIX: IS 456 REFERENCE CHARTS

Fig. 4: Modification Factor for Tension Reinforcement (Cl. 23.2.1)



$$f_s = 0.58\, f_y \frac{\text{Area of cross-section of steel required}}{\text{Area of cross-section of steel provided}}$$

FIG. 4 MODIFICATION FACTOR FOR TENSION REINFORCEMENT

Table 19: Design Shear Strength of Concrete, τc (Cl. 40.2.1)

Table 19 Design Shear Strength of Concrete, τ_c, N/mm²

(*Clauses* 40.2.1, 40.2.2, 40.3, 40.4, 40.5.3, 41.3.2, 41.3.3 *and* 41.4.3)

$100\frac{A_s}{bd}$	Concrete Grade					
	M 15	M 20	M 25	M 30	M 35	M 40 and above
(1)	(2)	(3)	(4)	(5)	(6)	(7)
≤0.15	0.28	0.28	0.29	0.29	0.29	0.30
0.25	0.35	0.36	0.36	0.37	0.37	0.38
0.50	0.46	0.48	0.49	0.50	0.50	0.51
0.75	0.54	0.56	0.57	0.59	0.59	0.60
1.00	0.60	0.62	0.64	0.66	0.67	0.68
1.25	0.64	0.67	0.70	0.71	0.73	0.74
1.50	0.68	0.72	0.74	0.76	0.78	0.79
1.75	0.71	0.75	0.78	0.80	0.82	0.84
2.00	0.71	0.79	0.82	0.84	0.86	0.88
2.25	0.71	0.81	0.85	0.88	0.90	0.92
2.50	0.71	0.82	0.88	0.91	0.93	0.95
2.75	0.71	0.82	0.90	0.94	0.96	0.98
3.00 and above	0.71	0.82	0.92	0.96	0.99	1.01

NOTE — The term A_s is the area of longitudinal tension reinforcement which continues at least one effective depth beyond the section being considered except at support where the full area of tension reinforcement may be used provided the detailing conforms to **26.2.2** and **26.2.3**

Table 20: Maximum Shear Stress, τc,max (Cl. 40.2.3)

Table 20 Maximum Shear Stress, $\tau_{c\,max}$, N/mm²

(*Clauses* 40.2.3, 40.2.3.1, 40.5.1 *and* 41.3.1)

Concrete Grade	M 15	M 20	M 25	M 30	M 35	M 40 and above
$\tau_{c\,max}$, N/mm²	2.5	2.8	3.1	3.5	3.7	4.0

FORMAL SIGN-OFF

DESIGN PREPARED BY

Name: _____

Designation: _____

Company: _____

Date: _____

Signature: _____

CHECKED BY

Name: _____

Designation: _____

Date: _____

Signature: _____